

UET University Peshawar

Abbottabad Campus



Lab Report No: 09(Weekend Lab)

Student Name: Muhammad Danial.

Registration Number: 24ABSWE0021

Class No: 21

Course: SE-305L Computer Communications and Networks

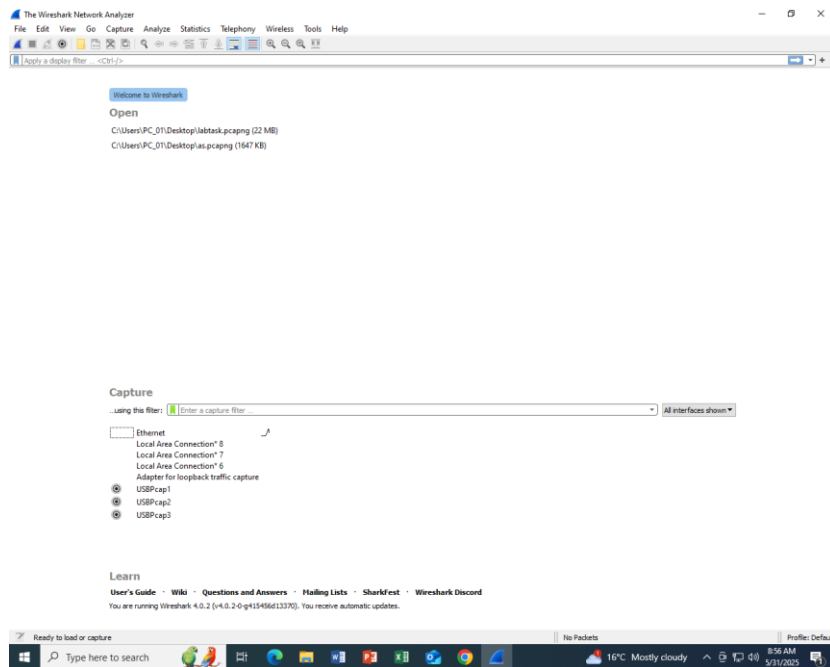
Instructor: Engr. Riasat Khan

Date: 18 June 2026

Department of Software Engineering
University of Engineering & Technology, Abbottabad Campus

Task 1: Install and Launch Wireshark

Task 2: Capture Live Traffic



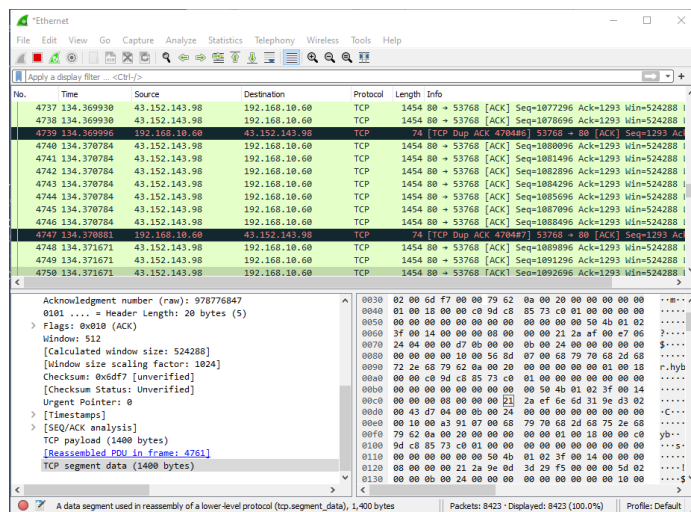
Q1. What happens in Wireshark when you start capturing?

When capturing starts, Wireshark begins recording all network packets passing through the selected network interface in real time. These packets appear in the packet list pane with color-coded protocol types.

Q2. Why did we visit websites during capture?

Visiting websites generates network traffic such as DNS queries, HTTP requests, and TLS handshakes. This traffic provides sample data for analysis during the lab.

Task 3: Analyze Protocols and Packet Details



Q3. Which protocol is used to resolve domain names?

DNS (Domain Name System) is used to resolve domain names into IP addresses.

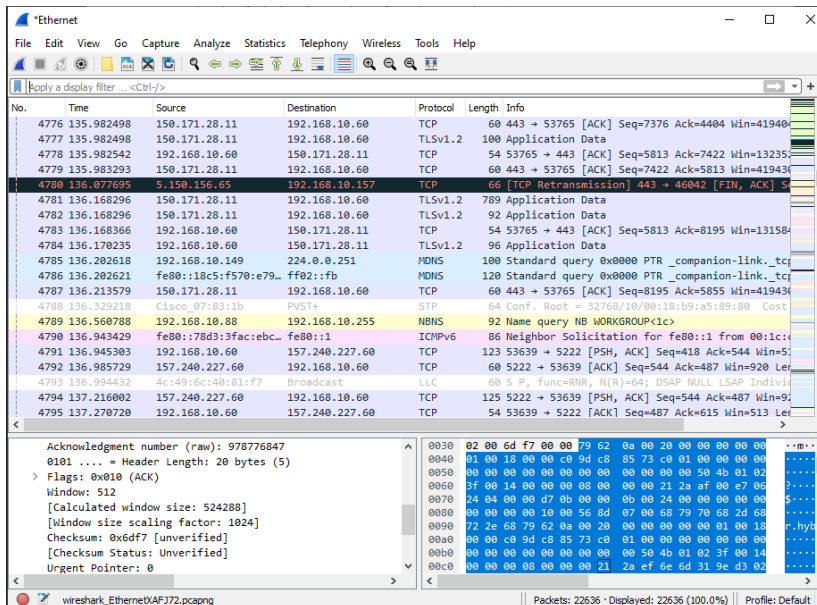
Q4. Which transport protocol does HTTP use?

HTTP uses the TCP (Transmission Control Protocol) transport layer protocol.

Q5. What information do the IP headers reveal?

IP headers reveal the source and destination IP addresses, the version of IP (IPv4 or IPv6), time to live (TTL), header length, and protocol used (e.g., TCP or UDP).

Task 4: Apply Basic Filters



Q6. What does the filter tcp display?

It displays only packets that use the TCP protocol, hiding all others such as UDP, ICMP, or ARP.

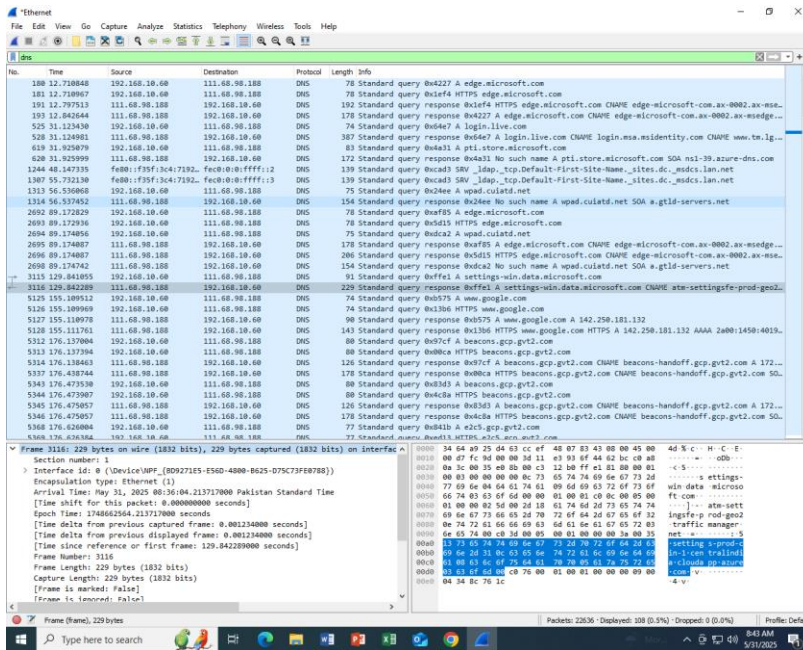
Q7. How can you isolate traffic to/from your own IP address?

Use the filter `ip.addr == YOUR_IP_ADDRESS`, replacing `YOUR_IP_ADDRESS` with your actual IP from an earlier packet.

Q8. What happens when a wrong or invalid filter is entered?

Wireshark shows a red background in the filter bar and does not apply the filter. No packets are displayed until a valid filter is entered.

Task 5: Follow a TCP Stream



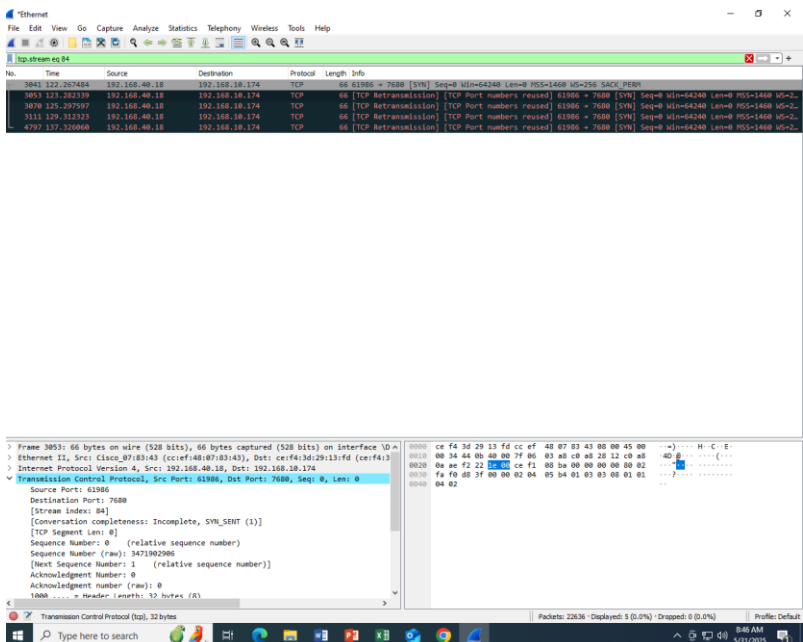
Q9. What information does "Follow TCP Stream" reveal?

It shows the complete communication (including request and response) between the client and server in a readable, chronological format. It helps track how data flowed across the TCP connection.

Q10. How can this feature be useful for troubleshooting application issues?

It helps in understanding how an application communicates, identifying failures like incomplete handshakes, errors in HTTP requests, or unexpected responses from the server.

Task 6: Analyze Packet Timing and Performance



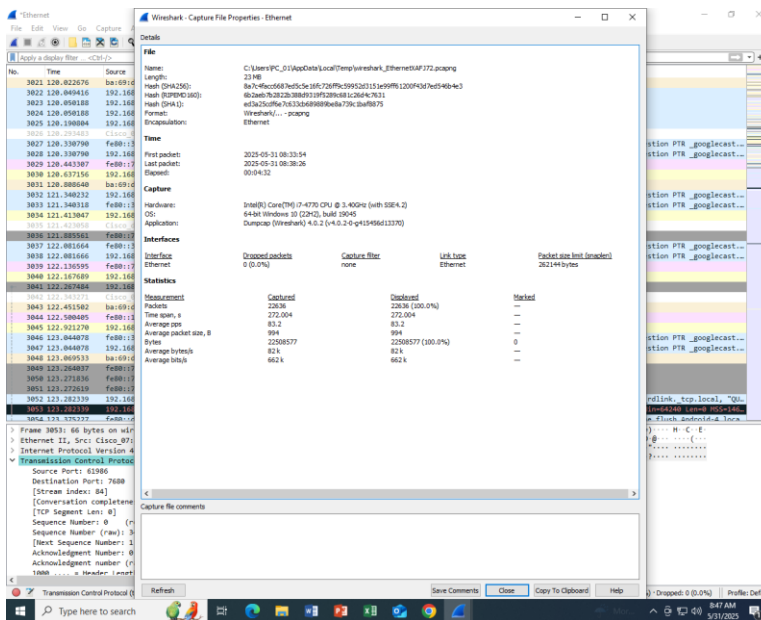
Q11. What does the Protocol Hierarchy window help you understand?

It shows a breakdown of all protocols observed in the capture session, along with their packet counts and percentage of total traffic, helping to understand protocol distribution and network activity.

Q12. How can IO Graphs be used to detect network spikes or issues?

IO Graphs plot traffic volume over time. Spikes can indicate heavy usage, potential attacks (like DDoS), or misconfigured systems. Comparing different protocols can reveal bottlenecks or latency.

Task 7: Save and Export Captures



Q13. Why is it important to save capture files?

Saving captures allows you to revisit and analyze the data later, share it with others for troubleshooting, or use it as evidence for security or performance issues.

Q14. What is the difference between .pcapng and .txt exports?

.pcapng is a binary format used by Wireshark to preserve full packet data and metadata.

.txt is a human-readable summary of the packet data, often used for documentation or simplified reviews.

Wireshark interface showing network traffic analysis. The top pane displays a list of captured packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The middle pane shows the details of the selected packet (Frame 3116), including Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Domain Name System (response). The bottom pane displays the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
188	12.718648	192.168.10.60	111.68.98.188	DNS	78	Standard query 0x4227 A edge.microsoft.com
189	12.718967	192.168.10.60	111.68.98.188	DNS	78	Standard query 0x42f4 HTTPS edge.microsoft.com
191	12.797513	111.68.98.188	192.168.10.60	DNS	192	Standard query response 0x42f4 HTTPS edge.microsoft.com CNAME edge-microsoft-com.ax-0002.ax-mse...
193	12.842644	111.68.98.188	192.168.10.60	DNS	178	Standard query response 0x4227 A edge.microsoft.com CNAME edge-microsoft-com.ax-0002.ax-mse...
525	31.122430	192.168.10.60	111.68.98.188	DNS	74	Standard query 0x4d07 A login.live.com
528	31.124981	111.68.98.188	192.168.10.60	DNS	387	Standard query response 0x4d07 A login.live.com CNAME login-msa-msidentity.com CNAME www.tn.lg...
619	31.925879	192.168.10.60	111.68.98.188	DNS	83	Standard query 0x4311 A pti.store.microsoft.com
620	31.925999	111.68.98.188	192.168.10.60	DNS	172	Standard query response 0x4311 No such name A pti.store.microsoft.com SOA nsl-39.azure-dns.com
1244	48.447335	fec0::f55f:3ca:7192::fec0::8:ffff::12	192.168.10.60	DNS	139	Standard query 0x4d03 SRV _ldap._tcp.Default-First-Site-Name._sites._msdcs.lan.net
1387	55.732130	fec0::f55f:3ca:7192::fec0::8:ffff::13	192.168.10.60	DNS	139	Standard query 0x4d03 SRV _ldap._tcp.Default-First-Site-Name._sites._msdcs.lan.net
1313	56.530608	192.168.10.60	111.68.98.188	DNS	75	Standard query 0x24ee A wpad.culatd.net
1314	56.537462	111.68.98.188	192.168.10.60	DNS	154	Standard query response 0x24ee No such name A wpad.culatd.net SOA a.gld-servers.net
2692	89.172829	192.168.10.60	111.68.98.188	DNS	78	Standard query 0x4f85 A edge.microsoft.com
2693	89.172936	192.168.10.60	111.68.98.188	DNS	78	Standard query 0x4f85 HTTPS edge.microsoft.com
2694	89.174066	192.168.10.60	111.68.98.188	DNS	75	Standard query 0x4dca2 A wpad.culatd.net
2695	89.174087	111.68.98.188	192.168.10.60	DNS	178	Standard query response 0x4f85 A edge.microsoft.com CNAME edge-microsoft-com.ax-0002.ax-mse...
2696	89.174087	111.68.98.188	192.168.10.60	DNS	206	Standard query response 0x4f85 HTTPS edge.microsoft.com CNAME edge-microsoft-com.ax-0002.ax-mse...
2698	89.174742	111.68.98.188	192.168.10.60	DNS	154	Standard query response 0x4dca2 No such name A wpad.culatd.net SOA a.gld-servers.net
3115	129.843955	192.168.10.60	111.68.98.188	DNS	93	Standard query 0xffe1 A settings-win.data.microsoft.com
3116	129.844289	111.68.98.188	192.168.10.60	DNS	228	Standard query response 0xffe1 A settings-win.data.microsoft.com CNAME ats-settingsfe-prod-geo2...
5125	155.109512	192.168.10.60	111.68.98.188	DNS	74	Standard query 0xb575 A www.google.com
5126	155.109519	192.168.10.60	111.68.98.188	DNS	74	Standard query 0xb1306 HTTPS www.google.com
5127	155.110978	111.68.98.188	192.168.10.60	DNS	96	Standard query response 0xb575 A www.google.com A 142.250.181.132
5128	155.111761	111.68.98.188	192.168.10.60	DNS	143	Standard query response 0xb1306 HTTPS www.google.com HTTPS A 142.250.181.132 AAAA 2a00:1458:4019...
5312	176.137804	192.168.10.60	111.68.98.188	DNS	80	Standard query 0xd97cf A beacons.gcp.gvt2.com
5313	176.137794	192.168.10.60	111.68.98.188	DNS	80	Standard query 0xb8ca HTTPS beacons.gcp.gvt2.com
5314	176.138463	111.68.98.188	192.168.10.60	DNS	126	Standard query response 0xd97cf A beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com A 172...
5337	176.438744	111.68.98.188	192.168.10.60	DNS	178	Standard query response 0xb8ca HTTPS beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com SO...
5343	176.473530	192.168.10.60	111.68.98.188	DNS	80	Standard query 0xb83d3 A beacons.gcp.gvt2.com
5344	176.473987	192.168.10.60	111.68.98.188	DNS	80	Standard query 0xb8ca HTTPS beacons.gcp.gvt2.com
5345	176.475057	111.68.98.188	192.168.10.60	DNS	126	Standard query response 0xb83d3 A beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com A 172...
5346	176.475057	111.68.98.188	192.168.10.60	DNS	178	Standard query response 0xb8ca HTTPS beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com SO...
5368	176.626884	192.168.10.60	111.68.98.188	DNS	77	Standard query 0xb413 A e2s5.gcp.gvt2.com
5369	176.626884	192.168.10.60	111.68.98.188	DNS	77	Standard query 0xb413 HTTPS e2s5.gcp.gvt2.com

Frame 3116: 228 bytes on wire (1832 bits), 228 bytes captured (1832 bits) on interface 0
Ethernet II, Src: Cisco-87:1b:43 (cc:ff:80:07:1b:43), Dst: Hewlett-Packard-25:04:16:35 (34:64:a9:15:04:16:35)
Internet Protocol Version 4, Src: 111.68.98.188, Dst: 192.168.10.60
User Datagram Protocol, Src Port: 53, Dst Port: 57483
Domain Name System (response)

Packets: 226/36 · Displayed: 108 (0.5%) · Dropped: 0 (0.0%) · Profile: Default